

Name: _____

Homework #08: General Purpose Input/Output

GPIO Read/Write & Bit-Specific Addressing

1. Please specify the LED (Red/Blue/Green) and SW (SW1/2) connected to the according pins on Port F on the TI LaunchPad. (Fill the table with Red, Blue, Green, SW1, and SW2)

PF4	PF3	PF2	PF1	PF0

2. If SW1 is not pressed by the user, what value is sensed by the microcontroller at the input pin? Assume that the internal pull-up resistor is being used. (Circle the correct answer)

1) High, 2) Low

3. Assume that you only want to read the SW1 input on the TI LaunchPad and no other bits. What bit-specific address would be used to read only that bit? (Assume APB.)

Answer: _____

4. Assume that you want to write to the 3 LEDs (Red LED, Blue LED, and Green LED) on the TI LaunchPad and no other bits. What bit-specific address would be used to access these 3 bits? (Assume APB.)

Answer: _____

5. Given the following code, what color is the tri-color LED on the TI LaunchPad right after this code is run?

```
LDR R1, =GPIO_PORT_LEDS ;loads the bit-specific address of all 3 LEDs
MOV R0, #2_00010
STR R0, [R1]
```

Answer: _____

6. For this problem, assume that V_{CC} in the image is equal to 3.3 [V]. You don't need to consider the small variations due to pull-up/down resistors, just use 3.3 V for high voltage (signal "1") and 0 V for low voltage (signal "0").

- a. What is the voltage at the input pin when the switch is not closed (the button is not pressed)? Circle the answer.

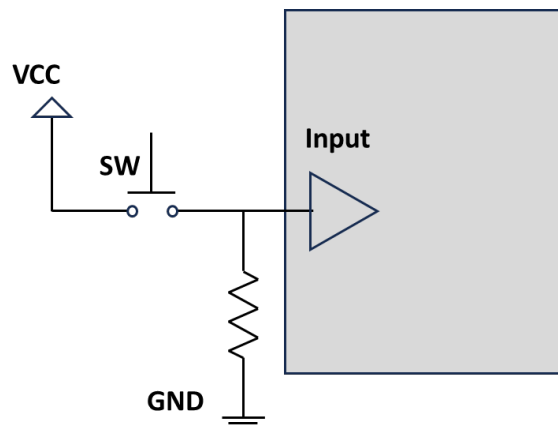
Answer: 3.3 V 0 V

- b. What is the voltage at the input pin when the switch is closed (the button is pressed)? Circle the answer

Answer: 3.3 V 0 V

- c. The resistor in the image below is called a _____ resistor. (Circle the correct answer.)

Answer: Pull-up Pull-down



GPIO Configuration: Step 1 – GPIO clock Activation Register

7. Write the code to turn on the clock for ports A, B, and D. You are not allowed to impact the clock enabled for the other ports. (Don't forget to add NOPs)

GPIO Configuration: Step 2 – GPIO Commit Register

8. Write the code to send the key (an immediate value) to Port B's Lock Register to let you have the priority to write to the port's Commit Register.

GPIO Configuration: Step 3 – GPIO Analog Mode select register

9. Write the code to select pin 4 of Port # as an analog input (as opposed to a digital input).

GPIO Configuration: Step 4 – GPIO Port Control Register

10. What is the address of Port C's Port Control Register

Answer: _____

GPIO Configuration: Step 5 – GPIO PIN Direction Register

11. Write the code to configure bits 0-3 for Port B as inputs and bits 4-7 as outputs.
(Remember that you are not allowed to change bits 8-31 of the configuration register.)

12. Write the code to configure Port to use four internal pull-up resistors for pin 0-3).

GPIO Configuration: Step 6 – GPIO Alternate Function Register

13. What is the address of Port A's Alternate Function Select Register?

Answer: _____

GPIO Configuration: Step 7 – GPIO DATA Pin Enable

14. Write the code to enable all 8 bits of Port B.